Budget-Robust Station Selection for Microtransit Under Demand Uncertainty

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Abstract

We extend a two-stage stochastic virtual bus stop (VBS) selection model to handle uncertain passenger demand within each scenario type (morning peak, midday, evening peak, weekend). Demand for each origin–destination pair (o, d) and scenario type s is modeled as lying in a capped-total-demand uncertainty set parameterized by lower and upper OD-specific bounds together with a per-scenario upper bound on aggregate demand. We derive the dual robust counterpart of the inner maximization, which remains linear and can be solved with the same MILP machinery as the nominal model. We compare three station-selection policies—*Nominal*, *Upper-Bound*, and *Robust Total-Demand-Cap*—on out-of-sample demand realizations and report average cost, high-percentile (P90/P95) cost, and the price of robustness.

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1 Introduction

Microtransit systems offer flexible, on-demand shared rides between a set of designated virtual bus stops (VBS). A key design decision is which candidate locations to build as VBS. Prior work formulates this as a two-stage stochastic mixed-integer program: in the first stage, build l stations and choose which $k \leq l$ are active in each scenario $s \in S$; in the second stage, assign passengers to origin–destination VBS pairs.

A limitation of the stochastic model is that it treats the demand q_{ods} within each scenario as a fixed, known quantity (typically the historical mean). In practice, demand fluctuates even within a scenario type: morning-peak demand on one day may differ substantially from the next. This motivates a *robust* extension where q_{ods} is uncertain within a bounded set.

The uncertainty model adopted here is designed to encode a direct operational statement: each OD pair has a plausible demand range, and the *total* demand within a scenario type should not exceed a calibrated upper bound. This produces a tractable robust counterpart while aligning more closely with the goal of limiting aggregate demand rather than normalized perturbation.

Even without explicitly modeling routing, this notion of robustness still has a clear interpretation. Under a nominal-demand optimization, the chosen station set may fit the average demand pattern well but perform poorly when demand shifts toward OD pairs that are less conveniently located relative to the selected stops. In those cases, realized passenger access cost can spike, and more requests may be pushed toward assignments that are close to the edge of the admissible walking-distance structure. The robust model reduces this sensitivity by favoring station configurations whose assignment performance remains more stable across plausible demand realizations. Thus, in the reduced model, robustness should be interpreted as protection against sudden deterioration in passenger walking/access quality, rather than as protection of routing efficiency itself.

2 Base Model

2.1 Decision variables

$y_j \in \{0,1\}$	build station j
$z_{js} \in \{0,1\}$	station j is active in scenario s
$x_{odjks} \in \{0,1\}$	assign OD pair (o,d) to origin VBS j, destination VBS k in scenario s

2.2 Nominal objective

The nominal two-stage objective minimizes expected passenger travel cost:

$$\min_{x,y,z} \sum_{s\in S} \sum_{(o,d)\in\Omega} \sum_{j,k\in J} q_{ods} \underbrace{\left(d_{oj}^{\text{orig}} + d_{dk}^{\text{dest}} + \lambda\, c_{jk}\right)}_{a_{odjks}} x_{odjks}, \tag{1}$$

where d_{oj}^{orig} is the walking distance from origin o to VBS j, d_{dk}^{dest} is the walking distance from VBS k to destination d, c_{jk} is the in-vehicle travel time between VBS j and k, and λ is a weight that converts routing time to the same unit as walking distance.

2.3 Constraints

$$\sum_{j,k} x_{odjks} = 1 \qquad \forall (o,d), s \tag{2}$$

$$x_{odjks} \le z_{js}, \quad x_{odjks} \le z_{ks} \qquad \forall (o,d), j, k, s \tag{3}$$

$$z_{js} \le y_j \qquad \forall j, s \tag{4}$$

$$\sum_{j} y_j = l, \quad \sum_{j} z_{js} = k \qquad \forall s \tag{5}$$

3 Full Routing Formulation Reference

To assess whether a robust counterpart remains tractable once routing is modeled explicitly, it is useful to keep the full March 18 route-capacitated formulation in view. This is the formulation that more directly reflects the JacqWang setting where demand affects both station assignment and route selection.

3.1 March 18 Route-Capacitated Model

Let

P	: set of passenger requests,
S	: set of scenarios,
R	: set of candidate routes,
J	: set of candidate stations,
$o(p)$, $d(p)$	: origin and destination of request p,
$t(p)$	: time associated with request p,
$\Delta(o(p))$	: admissible pickup stations within walking distance of $o(p)$,
$\Delta(d(p))$	: admissible dropoff stations within walking distance of $d(p)$,
τ^r	: travel-time or route-cost coefficient of route r,
ρ	: fixed cost of using a route,
Cap_r	: capacity of route r,
$\beta^r_{jkt\ell}$	: indicator that demand class (j,k,t) occupies capacity on segment ℓ of route r.

Decision variables:

$x_{pjks} \in \{0,1\}$: whether request p is assigned to pickup station j and dropoff station k in scenario s,
$\alpha^r_{jkts} \in \mathbb{Z}_+$: amount of class (j,k,t) loaded onto route r in scenario s,
$z_{js} \in \{0,1\}$: whether station j is used in scenario s,
$y_j \in \{0,1\}$: whether station j is globally selected/opened,
$\theta^r_{ts} \in \mathbb{Z}_+$: number of deployments of route r in time bucket t and scenario s.

Parameters linking demand and routing:

q_p : demand associated with request p,
c^{walk}_{pjk} : walking/access cost when request p is assigned to (j,k).

The objective is

$$\min \sum_{s\in S} \left[\sum_{p\in P} \sum_{j\in\Delta(o(p))} \sum_{k\in\Delta(d(p))} q_p\, c^{walk}_{pjk}\, x_{pjks} + \mu \sum_{r\in R} (\tau^r + \rho)\, \theta^{rs} \right]. \tag{6}$$

Subject to

$$\sum_{j\in\Delta(o(p))} \sum_{k\in\Delta(d(p))} x_{pjks} = 1, \qquad \forall p\in P,\ \forall s\in S, \tag{7}$$

$$x_{pjks} \le z_{js}, \qquad \forall p\in P,\ \forall j\in\Delta(o(p)),\ \forall k\in\Delta(d(p)),\ \forall s\in S, \tag{8}$$

$$x_{pjks} \le z_{ks}, \qquad \forall p\in P,\ \forall j\in\Delta(o(p)),\ \forall k\in\Delta(d(p)),\ \forall s\in S, \tag{9}$$

$$z_{js} \le y_j, \qquad \forall j\in J,\ \forall s\in S, \tag{10}$$

$$\sum_{j\in J} z_{js} = K, \qquad \forall s\in S, \tag{11}$$

$$\sum_{j\in J} y_j = L, \tag{12}$$

$$\sum_{p\in P:\, t(p)=t} q_p\, x_{pjks} \le \sum_{r\in R} \alpha^r_{jkts}, \qquad \forall j,k,t,\ \forall s\in S, \tag{13}$$

$$\sum_{j,k} \beta^r_{jkt\ell}\, \alpha^r_{jkts} \le Cap_r\, \theta^r_{ts}, \qquad \forall r\in R,\ \forall \ell\in V(r),\ \forall t,\ \forall s\in S. \tag{14}$$

3.2 Why robustness becomes harder here

In this formulation, uncertainty in q_p propagates through multiple coupled layers. It affects the walking-cost term in (6), it determines the left-hand side of the route-covering constraints (13), and that demand must then be covered by route-loading variables α^r_{jkts} subject to segment-capacity constraints (14). This is why a compact robust counterpart becomes much harder than in the reduced station-selection model: the uncertain demand does not just perturb objective coefficients, it changes the amount of flow that must be feasibly loaded onto routes. That structure is exactly the kind of setting where cutting-plane or scenario-generation approaches are more natural than a closed-form robust counterpart.

3.3 Routing-aware robust problem in min-max form

For the present application, both the globally built stations y and the scenario-active stations z should be fixed in advance. The operational variables x, α, and θ may still adapt after demand is realized. Under this interpretation, the robust routing-aware problem is best written in *min-max* form, where the inner objective is the optimal recourse value for the realized demand.

A natural stage split is therefore:

- first stage: choose globally built stations y and scenario-active stations z;
- adversary: choose a demand realization $q \in \mathcal{U}$;
- recourse: choose assignments x, route loading α, and route deployments θ.

Define the recourse value function

$$\Phi(y, z, q) := \min_{x,\alpha,\theta} \left\{ \sum_{p,j,k} q_p \, c^{walk}_{pjk} \, x_{pjk} + \mu \sum_{r,t} (\tau^r + \rho) \, \theta^r_t \right\} \tag{15}$$

subject to

$$\sum_{j \in \Delta(o(p))} \sum_{k \in \Delta(d(p))} x_{pjk} = 1 \qquad \forall p, \tag{16}$$

$$x_{pjk} \le z_j, \quad x_{pjk} \le z_k \qquad \forall p, j, k, \tag{17}$$

$$\sum_{p:\, t(p)=t} q_p \, x_{pjk} \le \sum_r \alpha^r_{jkt} \qquad \forall j, k, t, \tag{18}$$

$$\sum_{j,k} \beta^r_{jkt\ell} \, \alpha^r_{jkt} \le Cap_r \, \theta^r_t \qquad \forall r, \ell, t, \tag{19}$$

$$x_{pjk} \in \{0, 1\}, \ \alpha^r_{jkt} \in \mathbb{Z}_+, \ \theta^r_t \in \mathbb{Z}_+. \tag{20}$$

Then the robust design problem is

$$\min_{\substack{y,z: \\ z_j \le y_j \ \forall j, \\ \sum_j y_j = L, \ \sum_j z_j = K}} \max_{q \in \mathcal{U}} \Phi(y, z, q). \tag{21}$$

Here $\mathcal{U}$ is the chosen demand uncertainty set. The max operator reflects the fact that y and z are fixed before demand is known, while the value function $\Phi(y, z, q)$ captures the best routing-and-assignment response after the realization of q.

This definition of the recourse value function is not merely theoretical. It also describes how out-of-sample evaluation should be carried out if the robust model is to be interpreted consistently. Once a design (y, z) is fixed by the first-stage optimization, a realized order set or demand realization determines q, and the operational model should then resolve the assignment and route loading subproblem to compute the realized response. In other words, the out-of-sample simulation should evaluate $\Phi(y, z, q^{\text{test}})$ for each held-out realization, rather than evaluating the fixed design under a different or heuristic assignment rule. This is important because the robust design is only meaningful relative to the recourse mechanism it assumes.

3.4 Why this suggests a cutting-plane method

Formulation (21) suggests a decomposition-based solution rather than a compact robust counterpart. For fixed design decisions (y, z), the adversarial problem asks for a demand realization $q \in \mathcal{U}$ that maximizes the recourse value $\Phi(y, z, q)$. This naturally leads to a cutting-plane or column-and-constraint generation loop:

1. solve a master problem over (y, z) using only a current finite set of adversarial demand realizations;
2. for the resulting $(y^\star, z^\star)$, solve an adversarial separation problem to search for a new worst-case demand realization $q^\star \in \mathcal{U}$;
3. if such a realization is found, add it to the master and repeat.

Under this view, the master problem is a finite-scenario robust design model, while the separation problem is a worst-case-demand search coupled to the routing recourse model. This is the main conceptual reason to prefer cutting planes here: the adversarial effect of q is mediated through a large integer recourse value function, so exact dualization is unlikely to remain compact.

4 Uncertainty Set for Demand

4.1 Parameterization

For each OD pair (o, d) and scenario type s, assume we can estimate a lower bound $\underline{q}_{ods}$ and an upper bound $\overline{q}_{ods}$ on demand, and let $\overline{Q}_s$ denote an upper bound on *total* demand in scenario type s. The uncertainty set is

$$\mathcal{U}_s = \Big\{ q_s : \underline{q}_{ods} \le q_{ods} \le \overline{q}_{ods} \ \forall (o,d), \ \sum_{(o,d)\in\Omega} q_{ods} \le \overline{Q}_s \Big\}. \tag{22}$$

This uncertainty set says two things simultaneously:

- each OD pair must remain within its own historically calibrated range;
- the aggregate demand in the scenario cannot exceed $\overline{Q}_s$.

4.2 Interpretation of the total-demand cap

- If $\overline{Q}_s = \sum_{(o,d)} \underline{q}_{ods}$, then all OD demands are forced to remain at their lower bounds.
- If $\overline{Q}_s \ge \sum_{(o,d)} \overline{q}_{ods}$, then the total demand cap is inactive and the model reduces to independent box bounds on each OD pair.
- Intermediate values of $\overline{Q}_s$ allow some OD pairs to be high simultaneously, but only so long as total demand stays below the calibrated cap.

A larger $\overline{Q}_s$ for morning peak than for midday encodes the prior that peak periods experience larger systemwide demand surges.

4.3 Remark on integer demand

Observed demand is integer-valued, but the robust inner maximization may still be solved as a linear program. The key point is that the uncertainty set (22) is a bounded knapsack-type polytope. If the bounds $\underline{q}_{ods}$, $\overline{q}_{ods}$, and $\overline{Q}_s$ are integral, the inner LP has an optimal extreme point that is integral as well. Hence dropping integrality in the inner maximization does not change the worst-case value, even though the uncertainty set is written in continuous form.

4.4 Calibrating the uncertainty set from data

Group historical observations by scenario type $s \in \{\text{morning}, \text{midday}, \text{evening}, \text{weekend}\}$. For each OD pair (o, d) compute empirical quantiles:

$$\underline{q}_{ods} = Q_{0.10}\big(q_{od}^{\text{hist}} \mid s\big), \qquad \overline{q}_{ods} = Q_{0.90}\big(q_{od}^{\text{hist}} \mid s\big). \tag{23}$$

These quantiles are only an example; in practice they should be chosen to match the desired level of robustness and the amount of available data. If the lower quantile is set too high, the model can become infeasible or artificially tight because the total-demand cap must still exceed the sum of all lower bounds. For sparse OD pairs, one may regularize these estimates by pooling across nearby OD pairs, shrinking toward scenario-level averages, or imposing a minimum support threshold before assigning pair-specific bounds.

Next compute the realized total demand on each historical day:

$$Q_{s,t}^{\text{hist}} := \sum_{(o,d)\in\Omega} q_{ods,t}^{\text{hist}}, \tag{24}$$

where t indexes historical observations within scenario type s. Then set

$$\overline{Q}_s = Q_{0.90}\big(Q_{s,t}^{\text{hist}}\big) \quad \text{or} \quad \overline{Q}_s = Q_{0.95}\big(Q_{s,t}^{\text{hist}}\big). \tag{25}$$

Equation (25) is the direct data-driven analog of the modeling statement "total demand in scenario s should usually not exceed a high historical quantile." Choosing the 90th percentile gives a less conservative model, while the 95th or 99th percentile yields more protection against systemwide surges. One practical check is to verify that

$$\sum_{(o,d)} \underline{q}_{ods} \le \overline{Q}_s \le \sum_{(o,d)} \overline{q}_{ods},$$

and if this fails, adjust the quantile choices or rescale the OD-specific upper bounds to restore consistency.

4.5 Robust counterpart for the total-demand-cap set

Assume

$$\sum_{(o,d)} \underline{q}_{ods} \le \overline{Q}_s \le \sum_{(o,d)} \overline{q}_{ods} \qquad \forall s \in S,$$

so that the uncertainty set (22) is nonempty. Define the remaining demand budget after accounting for lower bounds:

$$B_s := \overline{Q}_s - \sum_{(o,d)} \underline{q}_{ods}.$$

Now substitute

$$q_{ods} = \underline{q}_{ods} + r_{ods}, \qquad \hat{q}_{ods} := \overline{q}_{ods} - \underline{q}_{ods}, \qquad 0 \le r_{ods} \le \hat{q}_{ods}.$$

Then, for fixed x, the inner worst-case problem becomes

$$\max_{r} \sum_{(o,d)} \big[\underline{q}_{ods} + r_{ods}\big] C_{ods}(x) \quad \text{s.t. } 0 \le r_{ods} \le \hat{q}_{ods},\ \sum_{(o,d)} r_{ods} \le B_s. \tag{26}$$

Separating the constant term gives

$$\sum_{(o,d)} \underline{q}_{ods} C_{ods}(x) + \max_{r} \left\{ \sum_{(o,d)} r_{ods} C_{ods}(x) : 0 \le r_{ods} \le \hat{q}_{ods},\ \sum_{(o,d)} r_{ods} \le B_s \right\}.$$

The second term is again a continuous knapsack problem, now with upper bounds $\hat{q}_{ods}$ on the excess-demand variables r_{ods}.

Proposition 1. For each scenario s, the inner maximization (26) has dual

$$\begin{aligned} &\max_{r} \sum_{(o,d)} r_{ods} C_{ods}(x) && \text{s.t. } 0 \le r_{ods} \le \hat{q}_{ods},\ \sum_{(o,d)} r_{ods} \le B_s \\ = &\min_{\alpha_s \ge 0,\, \beta_{ods} \ge 0} B_s\, \alpha_s + \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} && \text{s.t. } \alpha_s + \beta_{ods} \ge C_{ods}(x)\ \forall (o,d). \end{aligned} \tag{27}$$

Therefore the robust counterpart under the total-demand-cap uncertainty set is

$$\min_{x,y,z,\alpha,\beta} \sum_{s} \left[\sum_{(o,d)} \underline{q}_{ods}\, C_{ods}(x) + B_s\, \alpha_s + \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} \right] \tag{28}$$

subject to

$$\alpha_s + \beta_{ods} \ge C_{ods}(x) \qquad \forall (o,d),\ s, \tag{29}$$

plus the original constraints (2)–(5). Expanding $C_{ods}(x) = \sum_{j,k} a_{odjks}\, x_{odjks}$ gives the fully explicit linear form:

$$\min_{x,y,z,\alpha,\beta} \sum_{s} \sum_{(o,d)} \sum_{j,k} \underline{q}_{ods}\, a_{odjks}\, x_{odjks} + \sum_{s} B_s\, \alpha_s + \sum_{s} \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} \tag{30}$$

subject to

$$\alpha_s + \beta_{ods} \ge \sum_{j,k} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s, \tag{31}$$

plus (2)–(5).

5 Robust Objective

The robust model replaces the nominal objective with

$$\min_{x,y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \sum_{(o,d) \in \Omega} q_{ods}\, C_{ods}(x), \tag{32}$$

where the induced assignment cost $C_{ods}(x) = \sum_{j,k} a_{odjks}\, x_{odjks}$ depends only on the first-stage solution.

Using (22), this is equivalent to

$$\max_{q} \sum_{(o,d)} q_{ods}\, C_{ods}(x) \quad \text{s.t. } \underline{q}_{ods} \le q_{ods} \le \overline{q}_{ods},\ \sum_{(o,d)} q_{ods} \le \overline{Q}_s. \tag{33}$$

This is a continuous knapsack problem in the excess-demand variables $r_{ods} = q_{ods} - \underline{q}_{ods}$, and is most conveniently dualized.

6 Dual Robust Counterpart

The derivation in §4 shows that this inner maximization is equivalent to the dual formulation (27). Substituting back gives the robust counterpart (30), repeated here for convenience:

$$\min_{x,y,z,\alpha,\beta} \sum_{s} \sum_{(o,d)} \sum_{j,k} \underline{q}_{ods}\, a_{odjks}\, x_{odjks} + \sum_{s} B_s\, \alpha_s + \sum_{s} \sum_{(o,d)} \hat{q}_{ods}\, \beta_{ods} \tag{34}$$

subject to

$$\alpha_s + \beta_{ods} \ge \sum_{j,k} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s, \tag{35}$$

plus the original constraints (2)–(5). This is a mixed-integer linear program.

7 JuMP Implementation Sketch

The new data requirements are:

```
q_low[od, s]       # lower demand bound  (q_low_ods)
q_high[od, s]      # upper demand bound  (q_high_ods)
q_hat[od, s]       # = q_high - q_low   (q_hat_ods)
Q_cap[s]           # upper bound on total demand
B[s]               # = Q_cap[s] - sum(q_low[od,s] for od in OD)
a[od, j, k, s]     # assignment cost
```

New variables:

```
@variable(model, alpha[s in S] >= 0)
@variable(model, beta[od in OD, s in S] >= 0)
```

Robust objective:

```
@objective(model, Min,
    sum(q_low[od,s] * a[od,j,k,s] * x[od,j,k,s]
        for s in S, od in OD, j in J, k in J)
    + sum(B[s] * alpha[s] for s in S)
    + sum(q_hat[od,s] * beta[od,s] for od in OD, s in S)
)
```

Robust dual constraints:

```
@constraint(model, [od in OD, s in S],
    alpha[s] + beta[od,s] >=
        sum(a[od,j,k,s] * x[od,j,k,s] for j in J, k in J)
)
```

All original constraints (2)–(5) are unchanged.

8 Evaluation Experiment

8.1 Three model variants

We compare:

1. **Nominal model.** Uses $q_{ods} = \mathbb{E}[q_{ods}]$ (scenario-average historical demand).
2. **Upper-bound model.** Sets $q_{ods} = \overline{q}_{ods}$ for every OD pair. Maximally conservative within the OD-wise box bounds.
3. **Robust total-demand-cap model.** Uses uncertainty set $\mathcal{U}_s$ with $\overline{Q}_s$ calibrated from data via (25).

Each model produces a station selection $(y^\star, z^\star)$ and assignments $(x^\star)$ that are then *fixed* for out-of-sample evaluation.

8.2 Out-of-sample evaluation protocol

For each held-out demand realization q^{test}_{ods} (from a held-out date range or a parametric simulation), compute the realized cost:

$$\text{Cost}(x^\star, q^{\text{test}}) = \sum_{s,(o,d),j,k} q^{\text{test}}_{ods}\, a_{odjks}\, x^\star_{odjks}. \tag{36}$$

For the routing-aware robust model, this evaluation principle must be refined. The design variables (y, z) are fixed by the robust optimization, but the operational variables should be recomputed after the test realization is observed. Thus, for each held-out order set, the correct out-of-sample evaluation is to solve the recourse problem $\Phi(y^\star, z^\star, q^{\text{test}})$ and record the resulting assignment and routing cost. This makes the evaluation model consistent with the robust formulation itself and motivates an implementation path in which the simulator solves the routing-aware assignment subproblem conditional on fixed station decisions.

8.3 Reported metrics

For each model, report over the test set:

- Mean realized cost
- 90th and 95th percentile realized cost
- Worst-case realized cost
- Number of OD pairs poorly assigned (realized walking distance exceeding a threshold)
- Stability of station selection across demand realizations
- *Price of robustness*: $\dfrac{\text{Robust obj} - \text{Nominal obj}}{\text{Nominal obj}}$

8.4 Expected finding

The robust total-demand-cap model protects against systemwide demand surges without assuming that every OD pair simultaneously reaches its upper bound. It should give a *less conservative* solution than the full upper-bound model (lower in-sample objective, lower price of robustness) while improving high-percentile out-of-sample performance relative to the nominal model.

Formally:

$$\text{Nominal P90} \;\geq\; \text{Robust-Cap P90} \;\geq\; \text{Upper-Bound P90},$$

$$\text{Nominal mean} \;\leq\; \text{Robust-Cap mean} \;\leq\; \text{Upper-Bound mean}.$$

8.5 Experiment pipeline

The experiment is driven by scripts in `scripts/`:

1. `scripts/01_setup.jl`: reads `experiments/<name>/base.toml` and `experiments/<name>/sweep.toml`, generates per-job TOML configs.
2. `scripts/02_submit.sh`: SLURM array job template.
3. `scripts/03_run_job.jl`: loads config, runs the appropriate model, evaluates out-of-sample performance, saves `metrics.json`.
4. `scripts/aggregate.jl`: loads all `metrics.json` files, produces `analysis/results.csv` and prints summary statistics including the price of robustness.

The sweep in `experiments/2026-05-01_gamma_sweep/` can be repurposed to vary the calibration percentile used for $\overline{Q}_s$ (for example 90%, 95%, 99%) and the station budget $k \in \{6, 8, 10\}$.

9 Conclusion

We propose a robust extension of the VBS station selection model based on ODwise box bounds together with a scenario-level cap on total demand. The key insight is that the inner worst-case demand problem is a continuous knapsack in excess-demand variables, with a clean dual that yields a linear robust counterpart solvable by the same MILP solver as the nominal model. The parameters $\underline{q}_{ods}$, $\overline{q}_{ods}$, and $\overline{Q}_s$ have direct empirical interpretations and can be calibrated transparently from historical data. At the same time, the full routing formulation shows that a more operational notion of robustness would require protecting route loading and route cost under changing demand realizations, not just assignment quality. This points naturally to a next step in which robustness is imposed on the routing-aware model itself, likely through cutting-plane or scenario-generation methods rather than through a compact closed-form counterpart.

References

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